MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER PRACTICAL EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCECC200 -Electronics Service Technology

Duration: 1.5 hours Maximum Marks: 35

Practical Examination – Model Question Paper Instructions:

1. Demonstrate the given task.
2. A Viva session will follow the demonstration.
   (Minimum 3 questions from the Practical module)
3. Lab Report should cover all the subunits of the practical module.

Evaluation Criteria:

- **Viva:** 10 marks
- **Lab Record:** 10 marks
- **Demonstration:** 15 marks

Tasks: *(Choose any one from the list)*

1. Perform basic soldering of given discrete components on a PCB and inspect the solder joints.
2. Demonstrate desoldering of a faulty resistor/capacitor from a PCB using a desoldering pump or wick.
3. Demonstrate proper handling and use of soldering equipment and explain safety precautions.
4. Replace a faulty component on a given PCB using correct soldering and desoldering techniques.
5. Carry out fault finding in a DC power supply / SMPS unit using systematic testing methods.
6. Identify and troubleshoot faults in a LED bulb and suggest corrective measures.
7. Perform fault diagnosis on a table fan / ceiling fan and identify the defective part.
8. Troubleshoot an electric iron box and test the heating element and thermostat.
9. Perform continuity and insulation testing of a given household appliance.
10. Demonstrate the importance of earthing in protecting electronic equipment.
11. Identify and explain over-voltage and spike protection devices used in electronic circuits.

12. Demonstrate shielding techniques used to reduce noise and electromagnetic interference.
13. Inspect a PCB and suggest preventive measures to improve its reliability and lifespan.
14. Given a faulty PCB, perform complete servicing: inspection, testing, repair, and verification.
15. Carry out a complete soldering, testing, and fault analysis cycle on a provided circuit.